Course outline

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

Topic 5: BVPs and PDEs

- We typically use IVPs to study *dynamics* (time dependence)
- We typically use BVPs to study *spatial structures* (spatial dependence)
- PDEs give us the power to study both together

How Topic 5 builds on previous topics:

- We will use a solution method that reduces a PDE to a set of ODEs (cf. Topics 1–3)
- The PDE solution will typically be in the form of an infinite series (cf. Topic 2)
- The series are eigenfunction expansions, similar to Fourier series (Topic 4)

Topic 5: BVPs & PDEs

Leaning outcomes:

1. Understand how BVPs and PDEs arise in real-world applications
2. Be able to apply the separation of variables method to PDEs

Boundary Value Problems (BVPs)

Definition (Second-order, linear boundary value problem)

Consists of a 2nd order linear ODE

$$f_2(x)y''(x) + f_1(x)y'(x) + f_0(x)y(x) = g(x)$$

and two linear boundary conditions

$$a_1y'(x_1) + b_1y(x_1) = c_1 \qquad \text{and} \qquad a_2y'(x_2) + b_2\,y(x_2) = c_2$$

- If $c_1 = c_2 = 0$ then the boundary conditions are homogeneous
- If $c_1 = c_2 = 0$ and $g(x) = 0$ then the boundary value problem is homogeneous

Dirichlet and Neumann boundary conditions

- Dirichlet BCs fix the boundary values of the unknown function, e.g.,

$$y(0) = 0 \quad \text{and} \quad y(L) = 1$$

- Neumann BCs fix the derivatives, e.g.,

$$y'(0) = 0 \quad \text{and} \quad y'(L) = 0$$

- Robin BCs are weighted sums of Dirichlet and Neumann BCs, e.g.,

$$y(0) - y'(0) = 0 \quad \text{and} \quad 2\,y(L) + \pi\,y'(L) = 1$$

Exercise 5.1: Consider $y''(x) + y(x) = 0$ for $x \in (0, 2\pi)$

Which of the following BCs create a BVP with a unique solution?

1. $y(0) = 0$ and $y'(2\pi) = 0$
2. $y(0) = 0$ and $y(2\pi) = 0$
3. $y(0) = 1$ and $y(2\pi) = 0$

The general solution to $y''(x) + y(x) = 0$ is $y(x) = A\cos(x) + B\sin(x)$.

1. $y(0) = A\cos(0) + B\sin(0) = A = 0$ and $y'(2\pi) = -A\sin(2\pi) + B\cos(2\pi) = B = 0$ so we have the unique solution $y = 0$.

2. $y(0) = A = 0$ and $y(2\pi) = A\cos(2\pi) + B\sin(2\pi) = A = 0$ so we have the solution $y = B\sin(x)$, which is not unique.

3. $y(0) = A = 1$ and $y(2\pi) = A = 0$ and so we have no solution.

Structure of solutions

Unlike linear IVPs, linear BVPs are not guaranteed a unique solution (or even that a solution exists).

1. The complementary solutions to a homogeneous BVP form a vector space.
   - This vector space may be zero dimensional (i.e., zero is the only solution)
2. The solutions to a general linear boundary value problem are obtained by adding a single particular solution to the complementary solutions.
3. If no particular solution exists, then the general linear BVP has no solutions.

The solutions to BVPs play a key role in studying partial differential equations.

The Laplacian (Vector Calculus revision)

The Laplacian is a linear transformation that appears in many common PDEs. It is denoted by ∇^2 and its representation changes depending on the coordinate system:

1. 2D Cartesian

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

2. 3D Cartesian

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

3. Cylindrical

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2}$$

Partial differential equations (PDEs)

One of the differential operators is a partial derivative, and, hence, the unknown function is a function of multiple variables. Examples of some PDEs include:

- Laplace's equation

$$\nabla^2 u = 0$$

- Heat / diffusion equation

$$\dot{u} = \alpha \nabla^2 u$$

- Wave equation

$$\ddot{u} = c^2 \nabla^2 u$$

- Schrödinger's equation

$$i\hbar\dot{\psi} = -\frac{\hbar^2}{2m}\nabla^2\psi + V(x)\psi$$

- Liouville equation

$$\nabla^2 u + \mathrm{e}^{\lambda u} = 0$$

- Burgers equation

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2}$$